TCL Operators
&
Control Flow Statements

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Physical Design (PD)

Index

- Introduction to TCL
- Operators in TCL
    - Arithmetic Operators
    - Relational & Logical Operators
    - Bitwise & Ternary Operators
- Control Flow Statements
    - If-Else and Switch
    - For, While, Foreach Loops
- Practical Examples
- Summary Comparison

Operators : In TCL, Operators are special symbols or keywords used to evaluate expressions and perform tasks like:

1. Arithmetic Operators
2. Comparative Operators
3. Logical Operators
4. Bitwise Operators
5. Ternary Operators

Control Flow Statements : In TCL, Control Flow Statements are commands that control the flow of execution in a script, such as making decisions, repeating code, or choosing between options.

1. If, Elseif, Else
2. Switch case
3. For loop
4. While loop
5. Foreach loop

TCL Operators

1. Arithemetic operators:

+

-

*

/

%

**

2. Comparison operators:

==

!=

>

<

>=

<=

3. Logical operators :

&&

||

~

4. Bitwise operators:

&

|

^

<<

>>

4. Ternary Operator:

condition ? value_if_true : value_if_false

Control Structures

elseif statement :

```
if {cond} {
        body of if cond
} elseif {cond of the elseif} {
        body of elseif
} else {
        body of the else
}
```

Elseif statement contains if, else, elseif statements.

switch statement :

```
switch $var {
        "case1" {
                puts "print case1"
                }
        "case2" {
                puts "print case2"
                }
        "case3" {
                puts "print case3"
                }
        default {
                puts "default body"
                }
}
```

Control Structures

Foreach loop :

```
foreach var_name list {
    # code block
}
```

Var_name -- variable that will take each value from the list

list -- a space-separated list of values

code block -- commands to execute for each item

for loop :

```
for {initialization} {condition} {incr/decr} {
        body of the for loop
}
```

while loop :

```
set i 0
while {condition} {
        body of the while
loop
        incr/decr i
}
```

Example

Elseif loop :

Q. Write a TCL program that assigns a grade based on a variable score using if, elseif, and else statements. The grading criteria are: Grade A for scores 90 and above, Grade B for scores 75 to 89, Grade C for scores 60 to 74, Grade F for scores below 60 and display grade.

```
set score 75

if {$score >= 90} {
    puts "Grade: A"
} elseif {$score >= 75} {
    puts "Grade: B"
} elseif {$score >= 60} {
    puts "Grade: C"
} else {
    puts "Grade: F"
}
```

Output :

Grade: B

Example

Switch Statement :

Q. Write a TCL program that checks the value of a variable called fruit and prints a message based on the fruit name using a switch statement.

```
set fruit “orange"

switch $fruit {
        "apple" { puts “Apple is present"}
        "banana" { puts “Banana is present"}
        "orange" {puts “Orange is present"}
        default {       puts “Apple, Banana and orange is
not present"}
}
```

Output :

Orange is present

Example

For loop :

Q. write a tcl program to print all the prime numbers from 2 to 25.

```
for {set j 2} {$j <= 25} {incr j} {
    set isPrime 1
    for {set i 2} {$i < $j} {incr i} {
        if {$j % $i == 0} {
            set isPrime 0
            break
        }
    }
    if {$isPrime} {puts $j}
}
```

Output :

2
3
5
7
11
13
17
19
23

Example

While loop :

Q. write a tcl program to print all the numbers from 1 to 50 which is divisible by 4 and 6 using while loop.

```
set i 1
while {$i <= 50} {
        if {$i % 4 == 0 && $i % 6 == 0}
{
                puts $i
        }
        incr i
}
```

Output :

12
24
36
48

Example

Foreach loop :

Q. Iterating over a list of numbers and Add 10 to each number in the list.

```
set numbers {5 10 15 20}

foreach num $numbers {
    set result [expr {$num + 10}]
    puts "$num + 10 = $result"
}
```

Output :

5 + 10 = 15
10 + 10 = 20
15 + 10 = 25
20 + 10 = 30

Comparison table :

Statement	Purpose	Best Used For
if	Checks condition once	Simple decision-making
switch-case	Matches specific values	Handling multiple fixed options
for	Loops with initialization, condition, and increment	Loops with a known number of times or when you need to control the loop variable manually
while	Repeats while condition is true	Loops with unknown number of times
foreach	Loops through array/list	Easy element-by-element access

THANK YOU!

TCL Operators

1. Arithmetic Operators

Used to perform basic math operations like:

+ (Addition), - (Subtraction), * (Multiplication), / (Division), % (Modulus).

Example: set a [expr 10 + 5] # Output: 15.

2. Relational Operators

Used to compare two values and return true (1) or false (0):

<, >, <=, >=, ==, !=.

Example: expr 5 < 10 # Output: 1 (true).

3. Logical Operators

Used to combine multiple conditions:

&& (AND), || (OR), ! (NOT).

Example: expr {$a > 5 && $b < 10} # Output: 1 if both are true.

TCL Operators

4. Bitwise Operators

Used to perform bit-level operations:

& (AND), | (OR), ^ (XOR), ~ (NOT), << (Left Shift), >> (Right Shift).

Example: expr {5 & 3} # Output: 1.

5. Ternary Operator

Shorthand for if-else:

Syntax: condition ? value_if_true : value_if_false.

Example: expr {5 > 3 ? 5 : 3} # Output: 5.

Control Structures

Condition Control Structure:

1. **If :** Executes a block of code only if a specified condition is true.

   Elseif : Checks another condition if the previous if or elseif condition was false.

   Else : Executes a block of code if all preceding if and elseif conditions are false.

2. switch statement:

Compares a value against multiple cases and executes the matching block of code.

Loop Control Structure:

1. for loop:

Executes a block of code a fixed number of times based on initialization, condition, and increment.

2. while loop:

Keeps executing a block of code as long as the condition is true.

3. foreach loop:

Loops through each item in a list (or multiple lists) and runs the block of code for each element.